3B2v7.51c GML4.3.1 | RMMS : 1625 | Prod.Type:COM pp.1–13(col.fig.:13) | ED:Malini PAGN: dinesh SCAN: v4soft

ARTICLE IN PRESS

Available online at www.sciencedirect.com

SCIENCE DIRECT®

ELSEVIER

International Journal of Rock Mechanics & Mining Sciences ▮ (▮▮▮▮) ▮▮▮–▮▮▮

International Journal of Rock Mechanics and Mining Sciences

www.elsevier.com/locate/ijrmms

Interpreting hanging wall deformation in mines

D. Milne[a,*], R. Pakalnis[b], D. Grant[c], J. Sharma[a]

[a] *Civil and Geological Engineering Department, University of Saskatchewan, Saskatoon, Canada, SKS7N5A9*

[b] *Department of Mining & Mineral Processing Engineering, University of British Columbia, Vancouver, Canada*

[c] *Mining Engineer, Mount Isa Mines, Mount Isa, Queensland, Australia*

Accepted 2 May 2004

Abstract

The deformation behaviour of a hanging wall will follow distinct stages related to the changing material properties of the overall rock mass. This changing rock mass behaviour can be due to increasing underground opening size, blasting, changing stress conditions, as well as increasing opening exposure time. This paper examines the deformation behaviour of stope hanging walls due to changing opening geometry. The effect of time on deformation is observed, but is difficult to interpret. Field deformation data are presented from open stope hanging walls at Mount Isa Mine in Australia. Changing deformation response is related to increased opening spans recorded for two stopes. Some modelling studies are also presented which relate modelled elastic deformation near an opening surface to the opening geometry. Based on numerical modelling methods, it is difficult to predict the deformation of a hanging wall surface. This paper presents an approach for interpreting the rock mass behaviour based on the recorded deformation response of the hanging wall to continued mining.

© 2004 Published by Elsevier Ltd.

UNCORRECTED PROOF

1. Introduction

The deformation of an underground opening surface is dependent on many factors. For a completed mine opening with a static geometry, ongoing deformation is usually due to changing stress conditions, deteriorating ground conditions, nearby blasting or other effects related to opening exposure time. An underground opening with an active mining face, generally referred to as a stope, has a changing geometry and the deformation with time is predominantly due to increasing the size of the opening. This paper considers hanging wall deformation as the hanging wall size is increased due to continued mining activity, as studied by Milne [1]. Fig. 1 illustrates this situation.

Several factors influence hanging wall deformation with increased hanging wall size [2].

1. The volume of destressed rock in the hanging wall will increase causing deformation towards the stope opening.
2. The deformation modulus of the rock mass may decrease with continued deformation of the hanging wall, due to deterioration of the rock mass, leading to increased movement.
3. With further deformation, the rock mass may stop acting like a continuous linear elastic material.

The factors that influence hanging wall deformation are dependent on the distance to the supporting hanging wall abutments, or supporting walls. A term called the effective radius factor (ERF) is used to quantify the changing hanging wall geometry of a stope. The next section describes the development and application of the ERF term.

2. Effective radius factor (ERF)

The ERF term was developed to assess the relative stability of any arbitrary point on a hanging wall surface [3]. It has been modified from the hydraulic radius term used for empirically assessing the stability of underground surfaces [4], as can be seen from

*Corresponding author. Tel.: +306-966-5714; fax: +1-306-966-8593.

E-mail address: milned@engr.usask.ca (D. Milne).

1365-1609/$ - see front matter © 2004 Published by Elsevier Ltd.
doi:10.1016/j.ijrmms.2004.05.004

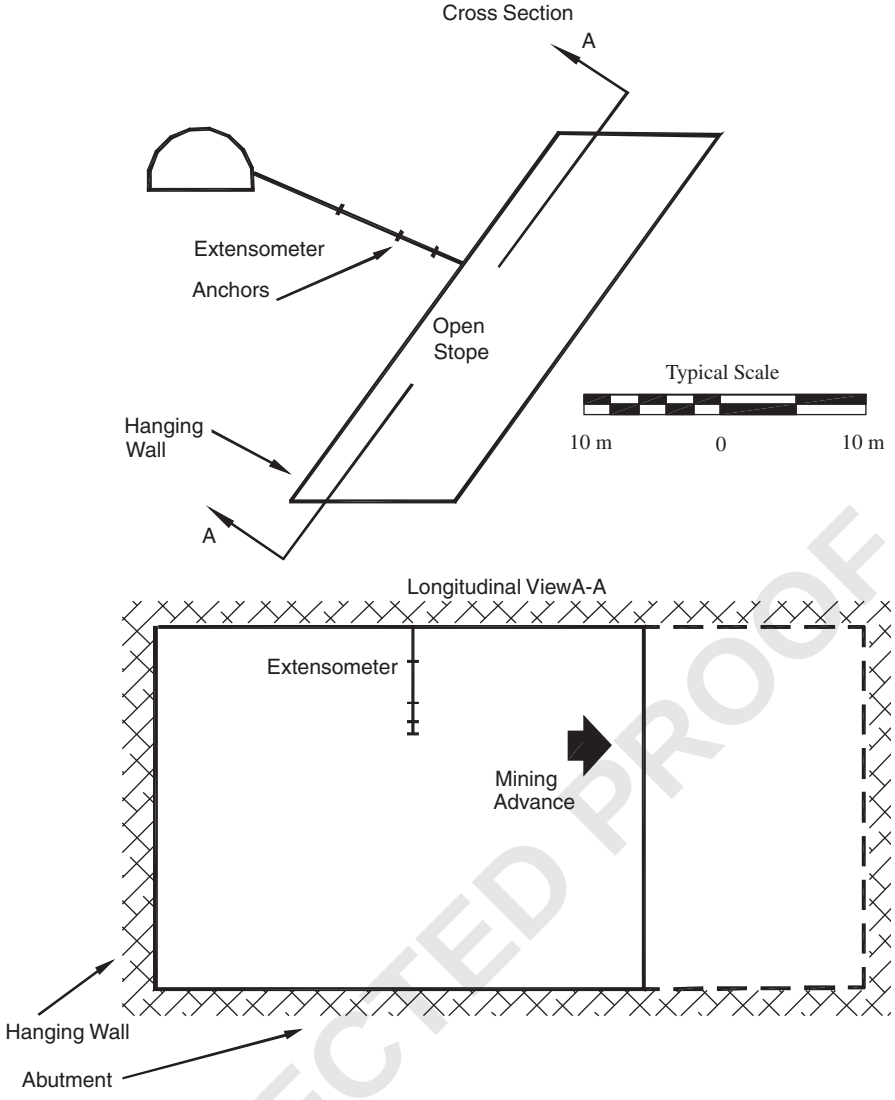


Fig. 1. General stope extensometer monitoring configuration (after Milne [1]).

$$\mathrm{HR} = \frac{ab}{2(a+b)} = \frac{2}{\left(\frac{1}{.5a}+\frac{1}{.5b}+\frac{1}{.5a}+\frac{1}{.5b}\right)} = \frac{0.5}{\frac{1}{4}\left(\frac{1}{.5a}+\frac{1}{.5b}+\frac{1}{.5a}+\frac{1}{.5b}\right)} \approx \mathrm{ERF}_{\mathrm{Max}} = \frac{0.5}{\frac{1}{n}\sum_{\theta=1}^{n}\left(\frac{1}{r_\theta}\right)}, \quad (1)$$

where a and b are the length and the width of an opening surface, respectively.

The hydraulic radius (HR) of a surface is simply the surface area divided by the perimeter. The maximum ERF value usually corresponds quite closely to the hydraulic radius. However, the relation between the two terms varies with the surface geometry. Table 1 shows the relation between HR and the maximum ERF as the surface length to width varies. In some circumstances the maximum ERF value can be used in place of hydraulic radius for empirical stability design, however, care must be taken to ensure the discrepancy between the two terms is not significant [3].

Table 1
Relation between hydraulic radius and max radius factor

Surface length to width ratio	$\mathrm{ERF}_{\mathrm{Max}}$ and HR relation
1:1	$\mathrm{ERF}_{\mathrm{Max}} \approx 1.11$ HR
2:1	$\mathrm{ERF}_{\mathrm{Max}} \approx 1.09$ HR
3:1	$\mathrm{ERF}_{\mathrm{Max}} \approx 1.05$ HR
5:1	$\mathrm{ERF}_{\mathrm{Max}} \approx 0.99$ HR
10:1	$\mathrm{ERF}_{\mathrm{Max}} \approx 0.86$ HR
∞:1 (Tunnel roof geometry)	$\mathrm{ERF}_{\mathrm{Max}} \approx 0.79$ HR

UNCORRECTED PROOF

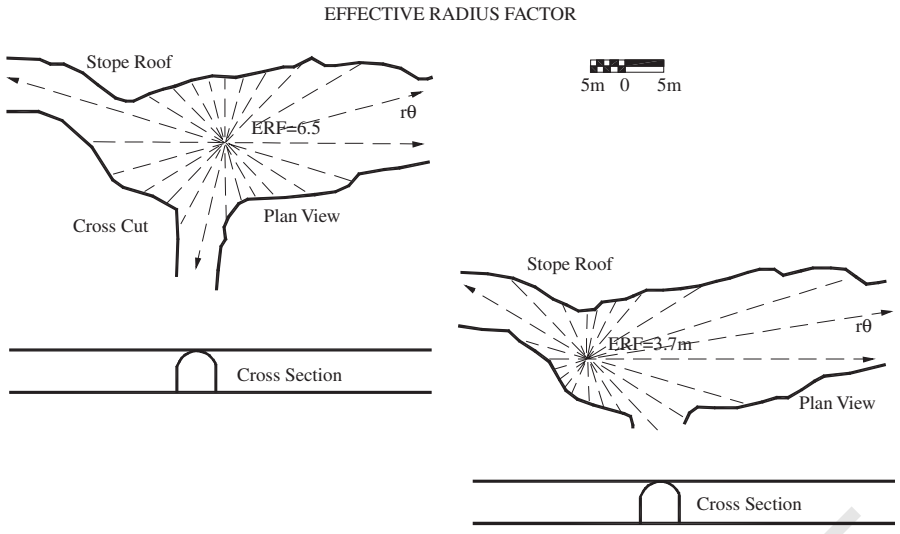


Fig. 2. An irregular stope roof showing the calculated effective radius factor value (after Milne [1]).

The ERF term can be calculated at any point on an opening surface, where r_θ is the distance to abutments measured at even angular increments (Eq. (1)). Hanging wall abutments or supporting walls are shown in Fig. 1. The calculated value of the ERF term converges to a relatively constant value when the angular measurements are less than 5° apart. Fig. 2 shows how this term can be calculated for an irregular opening surface. A program has been developed which will calculate the ERF value at any location on a surface [1]. The ERF value will equal zero at the edge or abutment of an opening and will be at a maximum at the centre of an opening [2].

3. Variation of modelled deformations with ERF values

A correlation between deformation and ERF values has been shown for computer modelled elastic deformation [1]. A 3-D (three-dimensional) boundary element program (Map3D, [5]) has been used to determine displacements normal to, and at a constant distance above an opening surface. The maximum elastic displacement normal to the opening occurs near the centre of the opening and the minimum displacement occurs at the surface abutments. Fig. 3 shows the general approach taken to relate measured or modelled deformation with the corresponding ERF value on or above an opening surface. The procedure can be broken into a number of steps (Fig. 3).

1. Choose a reference plane parallel to the excavation surface being studied.
2. Cut sections through this reference plane and determine elastic deformation of points on the section.
3. Project the points on the reference plane to the excavation surface. Determine ERF values for these points.
4. On each section cut, plot the elastic deformation of points on the reference plane corresponding to ERF values on the excavation surface.

Initially, deformations above a 50 m cubic opening were modelled 6.25 m above the surface. A uniaxial stress of 30 MPa was modelled with the stress oriented perpendicular to the surface with the measured deformations. Deformations were modelled 6.25 m above the cube on three measurement planes—diagonally across the surface, down the centre of the surface ($X = 25$ m) and along the quarter span ($X = 12.5$ m). Deformation along the three measurement planes were plotted against the ERF values at the modelled points on the plane. The results show a significantly linear relation between ERF values and deformation for the three combined measurement planes (Fig. 4). Deformations near the abutments diverged from linearity with the ERF values, probably due to the component of downward movement caused by the compression of the abutments. ERF values are zero at the abutment, or edges, of an opening. The minimum abutment compression would occur near the opening corners and the maximum abutment compression would occur at a point on the abutment furthest from the corners.

A second model was run with a hydrostatic stress field, to better match mining conditions. This run was made to determine if a linear ERF to deformation relation would still be obtained if stresses parallel to the opening were considered. Deformations were modelled at the same locations (Fig. 5). The magnitude of deformation was slightly lower in the hydrostatic stress field but the near linear relation between the ERF value

UNCORRECTED PROOF

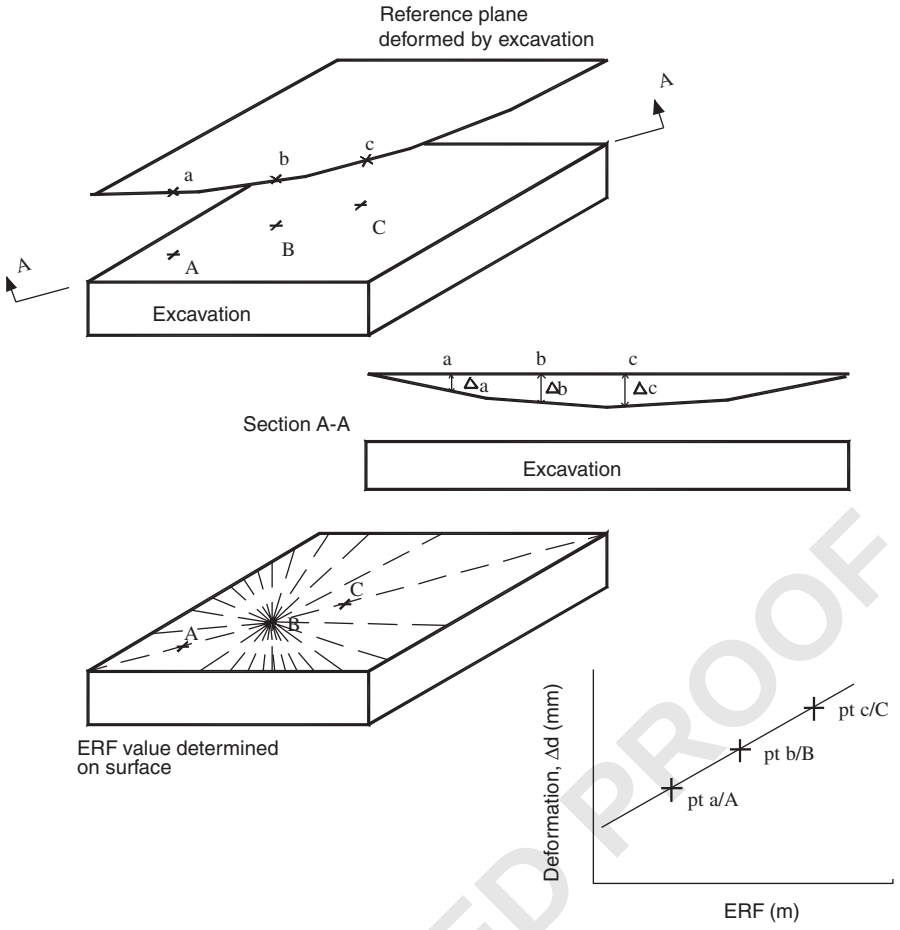


Fig. 3. General procedure for comparing modelled elastic displacements above a surface to corresponding ERF values on the surface (after Milne [1]).

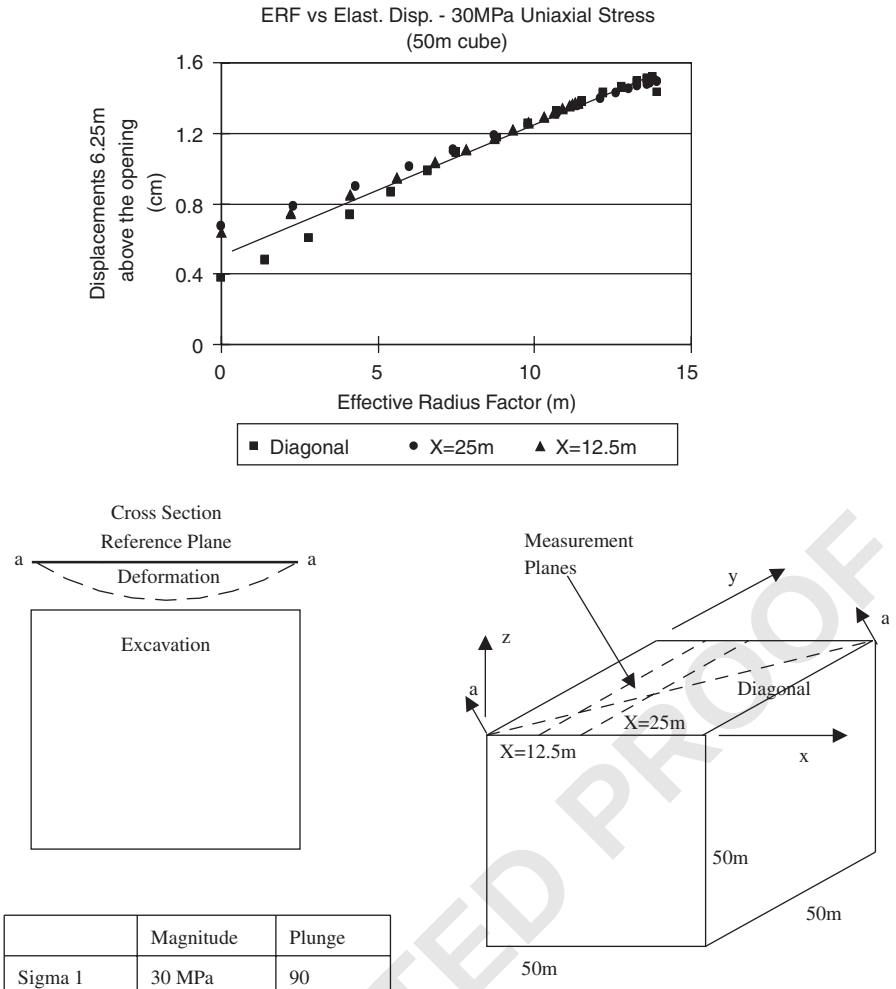
and the modelled elastic deformation was still apparent (Fig. 5). The slope of the elastic deformation versus ERF value was flatter for the case with the hydrostatic stress state.

The two stress states modelled represent a range in the stress ratio K from $K = 1.0$ for the hydrostatic stress state to K equal to infinity for the uniaxial stress state. A third model was run of an irregular shaped tabular opening in a hydrostatic stress field. The irregular opening was approximately 210 m long by 120 m wide by 5 m high (Fig. 6). A relatively linear relation was observed for elastic displacements versus ERF values above the single irregular tabular opening.

In all models, the ERF to deformation graph deviated from linearity near the abutments, where the ERF value is close to zero. Abutment compression cannot be accounted for by using the simple ERF factor, which is always equal to zero at abutments. Other model runs were conducted on various simple shapes of single openings in varied stress fields. The deformation of surfaces at varied distances above openings were plotted versus the corresponding ERF value on the surface and in all cases a near linear relation was obtained which diverged from linearity only near the abutments [1]. The magnitude of deformation varied with the stress field and rock properties modelled. The near linear relation between deformation and the ERF value was observed in all model runs [1]. Recent work by Pascoe and Oddie [6] also reported a direct relation between the ERF value and corresponding modelled elastic deformation.

The magnitude of modelled deformation versus corresponding ERF values is only of academic interest since elastic deformations can easily be estimated from computer models. The consistent near linear relation between increasing ERF values and modelled elastic deformation is of interest as a means of interpreting observed field deformations. It is proposed that when a linear relation between the ERF value and the corresponding deformation above the surface is obtained, the rock mass is deforming elastically. When this relation between deformation and the ERF value exceeds linearity, the rock mass is no longer deforming in a simple elastic mode. The next section looks at field deformations, which are much more difficult to model realistically.

UNCORRECTED PROOF



	Magnitude	Plunge
Sigma 1	30 MPa	90
Sigma 2	0	0
Sigma 3	0	0
E (GPa)	60	
Poissons	0.25	

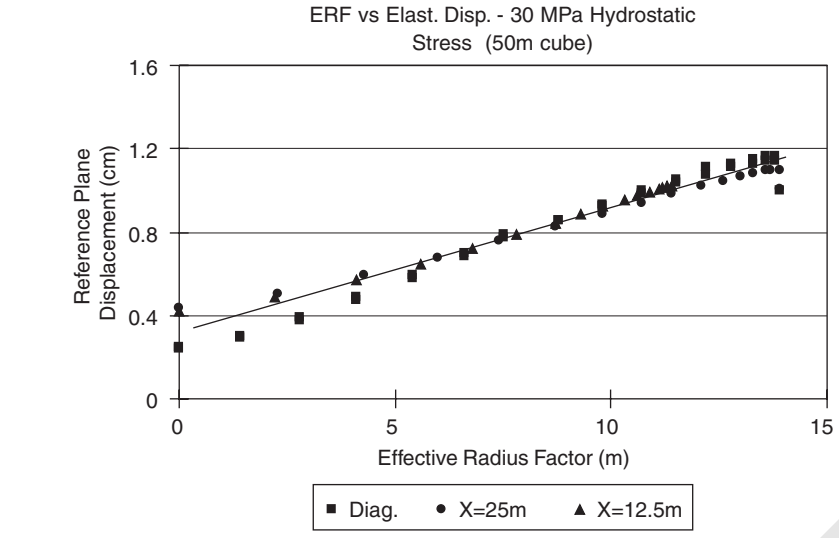
Fig. 4. Elastic deformation modelled 6.25 m above a 10 m cubic opening in a uniaxial stress field, plotted against the corresponding ERF values (after Milne [1]).

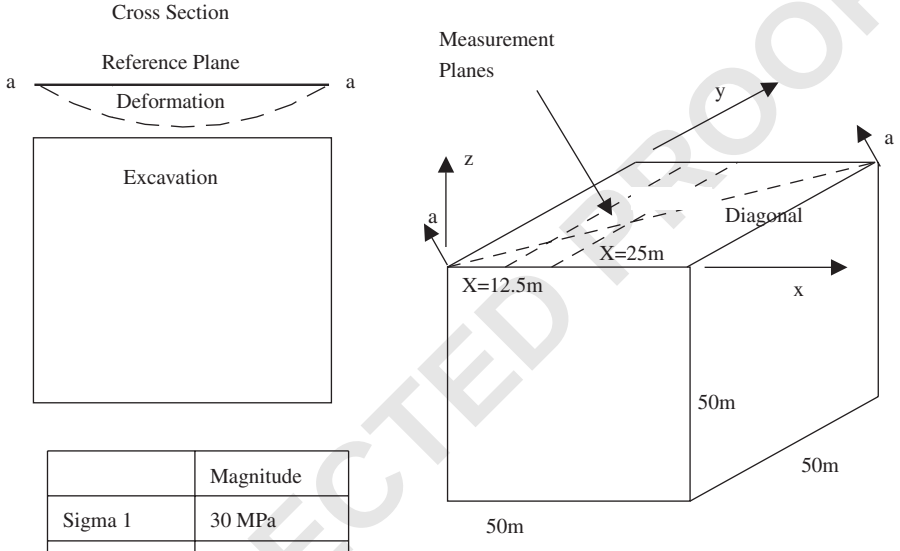
UNCORRECTED PROOF

4. Approach to field data analysis

Hanging wall deformation has not been analysed with respect to time because of the changing conditions due to continued mining. For each of the case histories, data analysis starts with the 'mining by' or the undercutting of the extensometer anchors. The term 'mining by' an instrument location refers to the point at which the mining front reaches the longitudinal projection of the instrument location. Fig. 7 shows a stope with a hanging wall extensometer immediately before and after mining by the instrument location. Stress related deformation of instruments, while located in stope abutments, is ignored (prior to mining by). The position of each instrument and anchor location is projected onto a true longitudinal view of the stope hanging wall. With each mining step after the extensometer has been mined by, the ERF value is calculated for each anchor location with respect to the new abutments. For each mining step, deformation readings for each anchor are determined. Some time-dependent creep may be occurring, however, this is usually not significant for the short time periods between stope blasts, unless the anchor position is close to failure.

The calculated ERF value for each extensometer anchor and mining step is then plotted against total anchor deformation. Deformation is expressed as total movement from the collar. Based on these graphs, insight into the rock mass response to mining as well as the effect of future mining can be made. The previous





	Magnitude
Sigma 1	30 MPa
Sigma 2	30 MPa
Sigma 3	30 MPa
E (GPa)	60
Poissons	0.25

Fig. 5. Elastic deformation modelled 6.25 m above a 10 m cubic opening in a hydrostatic stress field, plotted against the corresponding ERF values (after Milne [1]).

UNCORRECTED PROOF

section has shown how the effective radius factor of a point on an opening can be related to the elastic deformation of the point above the surface. The next section will look at deformation collected in the field, and show how it relates to mining geometry.

5. Instrumented case histories—Mount Isa Mine

Data from several instrumented stoping blocks were collected from the Lead Mine at Mount Isa and two of these case histories are presented [1]. The Mount Isa Lead Mine is located in the valley of the Leichhardt River in northwestern Queensland, Australia. The ore body consists of deposits of copper and silver–lead–zinc in parallel seams of a shale formation about 1000 m thick.

The lead–silver–zinc ore body consists of multiple parallel beds dipping approximately 70° to the west. The ore body hanging walls are bedded shales of varied strength, with bedding oriented parallel to the hanging wall. Hanging wall stability and dilution are a priority

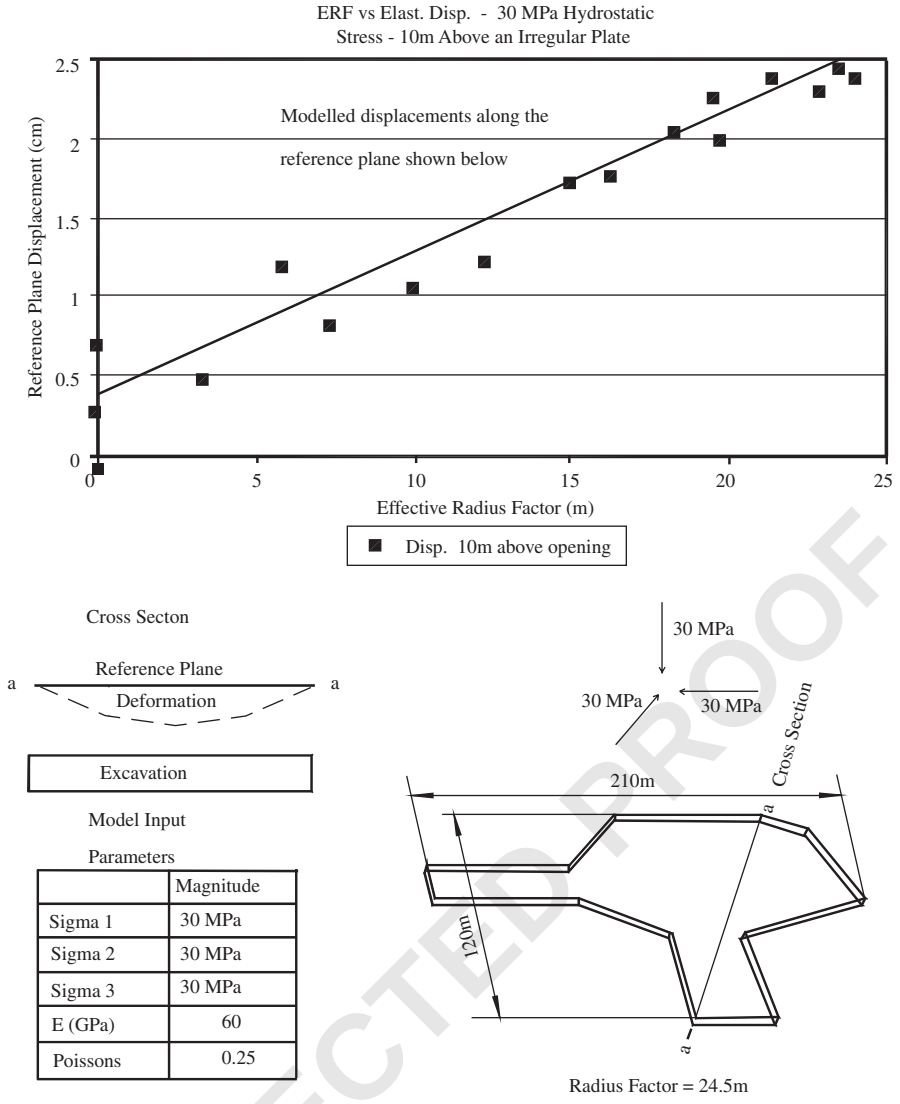


Fig. 6. Elastic deformation modelled 10 m above a 5 m high irregular tabular opening, plotted against the corresponding ERF value (after Milne [1]).

and a great deal of effort and research has gone into studying and instrumenting stope hanging walls.

In general, the ore body hanging wall has three predominant discontinuity sets, described by Windsor et al. [7]:

- Bedding planes are very continuous and weak with little or no cohesive strength and angles of friction typically in the order of 10°. The bedding planes are parallel and form the hanging wall surfaces.
- Steep north and south dipping joints which are continuous up dip, but tend to terminate on bedding.
- Steep east dipping joints, often with calcite infilling.

The major controlling rock mass properties for the hanging walls are the spacing, strength and continuity of the bedding planes.

The main mining methods used in the silver–lead–zinc ore body are open stoping and bench stoping. With these mining methods, large open hanging walls may be exposed which leads to a corresponding concern for the control of hanging wall dilution.

A substantial amount of instrumentation data on hanging wall behaviour has been collected at the mine. The closely spaced parallel ore bodies have made access for extensometer instrumentation relatively simple. Hanging wall deformation data have been obtained from several stopes, and two case histories are given in the next section.

6. 80G3 Bench stope instrumentation

Bench stope mining operations have been underway for over 15 years at the Mount Isa Lead Mine. A bench

UNCORRECTED PROOF

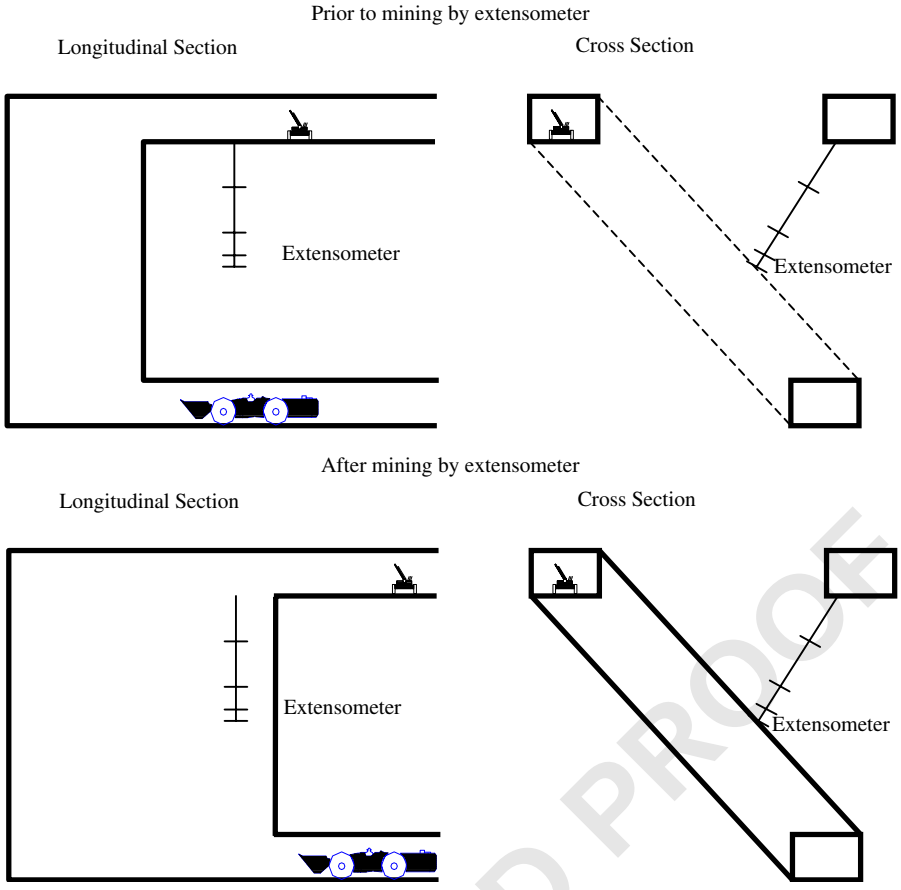


Fig. 7. General mining configurations showing a hanging wall extensometer before and after "mining by" of the stope (after Milne [1]).

stope is created with drifts driven at the top and bottom of the ore block along its entire strike length. Blasting is conducted on the upper overcut bench and mucking is done from the lower bench. A detailed description of the bench stoping method is given by Tucker et al. [8].

An extensive hanging wall instrumentation program was implemented for the mining of the initial panel stope in the 8 ore body at Mount Isa Lead Mine (Fig. 8). The 80G3 panel stope, at a depth of about 870 m, was the first panel stope in the 8 ore body and was considered a "trial stope". Stope 80G3 was mined at a depth of 870 m and the virgin stresses in the area were estimated as follows, based on internal MIM documents and work by Bywater et al. [9]:

$\sigma_1 = 0.040 * \text{depth} = 35$ MPa
(Oriented perpendicular to the body),
$\sigma_1 = 0.023 * \text{depth} = 20$ MPa
(Oriented sub-horizontal along the strike of the ore body),
$\sigma_1 = 0.020 * \text{depth} = 17$ MPa
(oriented down dip of the ore body).

Previous mining in the 8 ore body consisted of sub-level open stoping with relatively small hanging wall strike lengths exposed. In order to determine the behaviour of the large zone 8 ore body hanging wall, an instrumentation program consisting of 6 extensometers was implemented. The information for this summary was taken from internal mine documents.

The final open stope strike length was 71 m and the up dip bench dimension was 50 m. No significant hanging wall stability problems were noted during the mining of this stope.

Mapping was done in the area and indicated bedding plane breaks in the immediate 5 m of hanging wall were in the order of 4 to 5 breaks per metre. Based on ground condition, blasting and stress, a hanging wall stability assessment was made which indicated good hanging wall conditions [10]. Six extensometers on 4 rings were installed to monitor the behaviour of this bench stope hanging wall. The extensometer rings were 5 m apart and were located in the first-half of the stope excavated (Fig. 8). The bearing and dip of each extensometer hole was surveyed every 5 m down the hole. The hole and anchor locations are shown in long and cross section in Figs. 8 and 9. The monitoring of the hanging wall was

UNCORRECTED PROOF

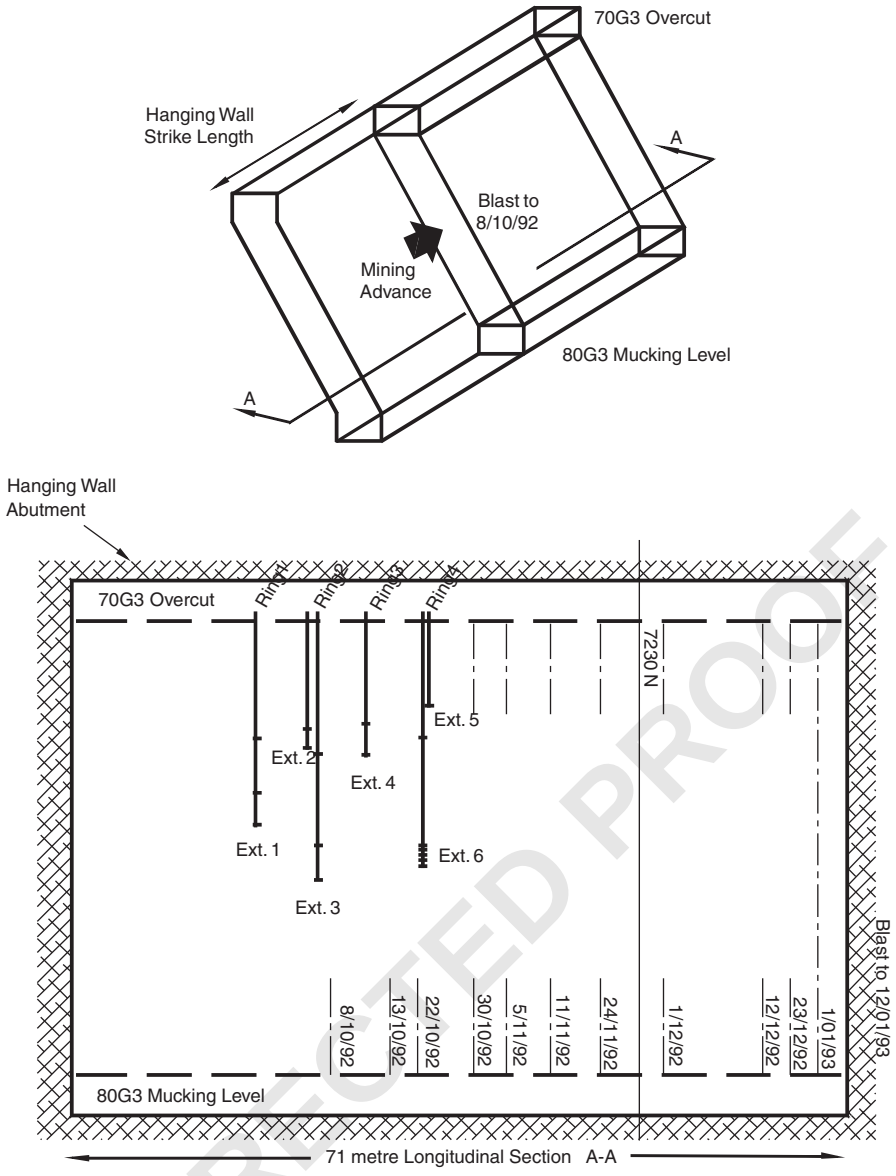


Fig. 8. Mount Isa Mine, stope 80G3 longitudinal view showing extensometer locations and blasting steps in longitudinal and isometric views (after Milne [1]).

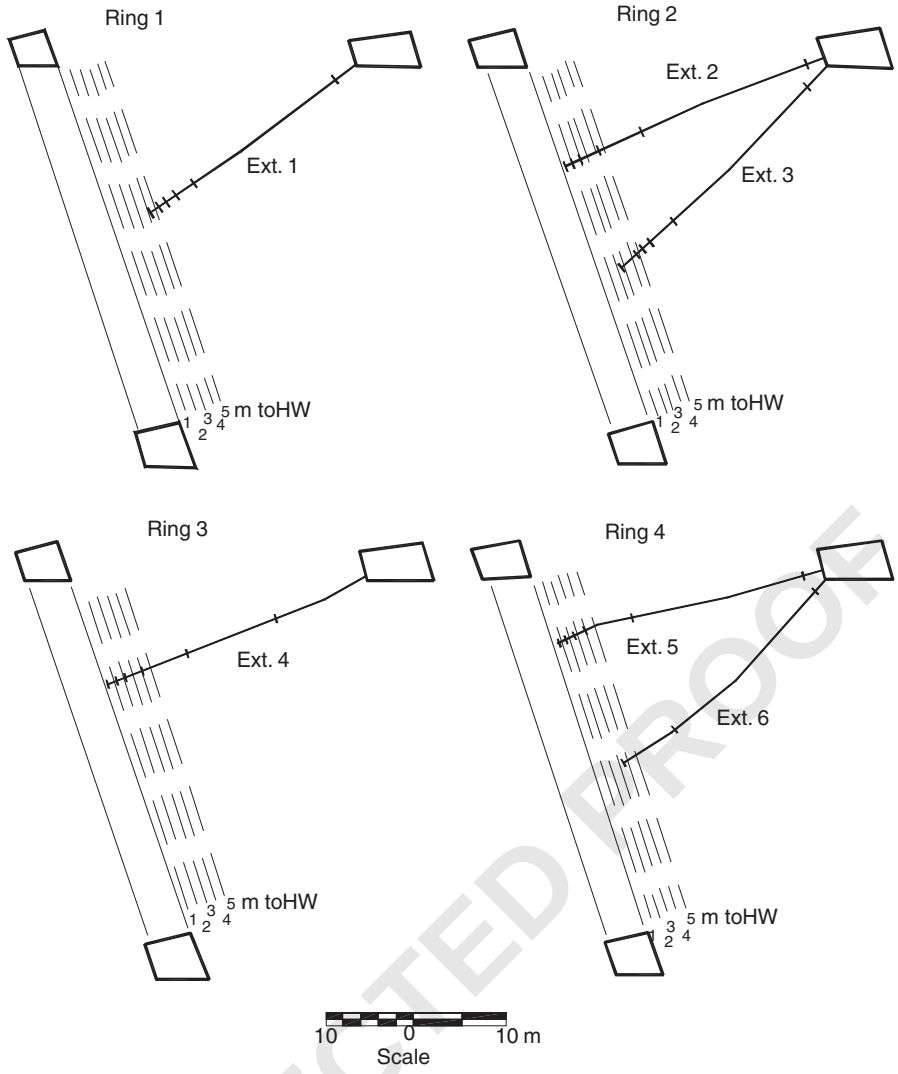
divided into 1 m wide bands and extensometer anchors were installed to monitor movement at the centre of these bands. The bands were defined as a distance from the hanging wall (Fig. 9).

The measured extensometer deformation has been related to the extensometer anchor distance from the supporting abutment, as shown in the longitudinal view (Fig. 8). With each blast and new stope limits, the ERF value at each extensometer anchor location has been assessed. Graphs of the movement from the collar to each anchor were analysed with respect to the change in the ERF value with mining.

Fig. 10 shows the variation in movement between the collar and anchors for 3 extensometers (extensometers 1, 3 and 5), with all the anchors located at approximately 5.5 m from the open hanging wall. In each case, the anchors showed a similar linear increase in movement with increasing ERF values suggesting movement is within the elastic range of the rock mass. The differences in the magnitude of deformation with ERF values is not large and may be due to variations in the actual distance between the anchors and blasted hanging wall surface.

This instrumentation program gave detailed data on the deformation behaviour of the immediate hanging wall of the 80G3 panel stope. Fig. 11 shows movement of extensometer anchors at 3 different distances from the open hanging wall. For the anchor at a distance of 0.5 m from the hanging wall contact, movement between the collar and anchor increases linearly to an ERF value of about 11 m, suggesting an elastic rock mass response

UNCORRECTED PROOF



Note: Extensometer positions are based on down hole surveys.

Fig. 9. Mount Isa Mine, stope 80G3 cross sections showing extensometer locations (after Milne [1]).

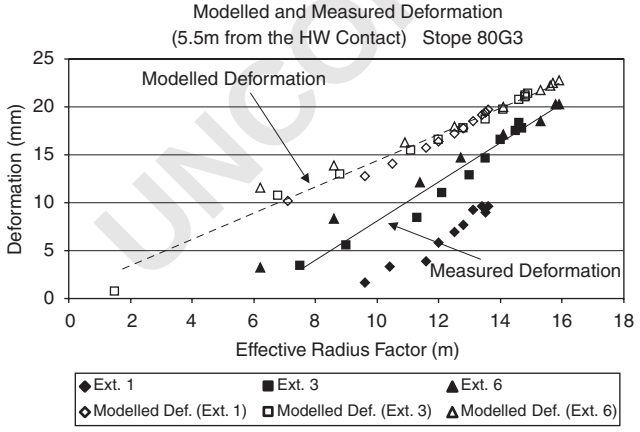


Fig. 10. Measured deformation versus corresppoinding effective radius factor values measured 5.5 m from the hanging wall contact (80G3 bench stope, Mount Isa Mines).

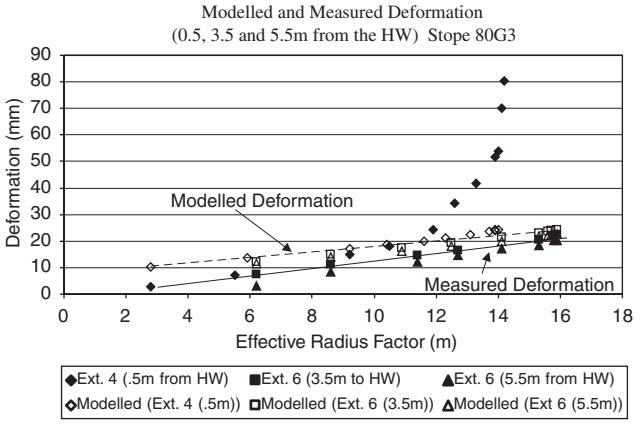


Fig. 11. Measured deformation versus corresponding effective radius factor values measured 0.5, 3.5 and 5.5 m from the hanging wall contact 80G3 bench stope, Mount Isa Mines.

UNCORRECTED PROOF

to continued mining. After an ERF of about 11 m, the deformation increases rapidly with increasing ERF values, suggesting a less stable, non-elastic response to continued mining. At 3.5 and 5.5 m from the open stope there is a linear increase in movement between the anchors and collar with increasing ERF values and the magnitude of movement is less than for the anchor at 0.5 m. At this greater distance from the open hanging wall, it appears that the rock mass is undergoing elastic deformation.

7. Stope 80G3 modelling

Modelling of this stoping block was conducted to determine if the recorded movements correspond to linear elastic deformation. A 3-D boundary element program (Map3D [5]) was again used. The modelled deformation perpendicular to the hanging wall surface was recorded for the mining geometries defined by the stope blasts shown in Fig. 8.

Virgin stress conditions were input for the 3-D computer modelling, with the maximum principal stress oriented normal to the ore body with a magnitude of 35 MPa. For the deformation properties of the rock mass, a value of 0.2 was set for Poisson's ratio. Young's modulus, based on an estimated RMR classification value of 70, was estimated at 40 GPa using Eq. (2) from Bieniawski [11].

$$E = 2\mathrm{RMR} - 100. \quad (2)$$

Modelling results are shown with the monitoring data in Fig. 10 for anchor depths at 5.5 m from the open hanging wall. The modelled deformation for the 3 extensometer anchors, all 5.5 m from the hanging wall, showed a consistent linear trend. The recorded field deformation followed a linear trend for each extensometer but the field data from the 3 extensometers did not lie on the same line. Extensometer 1 showed the least deformation for a given ERF value while extensometer 6 showed the greatest deformation. This variation in deformation could be due to minor differences in the actual distance between the anchor location and the open stope hanging wall.

Fig. 11 shows modelled and measured deformation 0.5, 3.5 and 5.5 m from the open stope hanging wall. Initially, a similar linear trend is obtained between the modelled and recorded hanging wall deformations when plotted against the extensometer anchor ERF values. The extensometer anchor 0.5 m from the hanging wall showed a rapid increase in deformation after an ERF value of 11 m, and ceased to correspond to the modelled deformation estimates. The modelling does indicate the rock mass was behaving as a linear elastic material, except within 0.5 m of the hanging wall when an ERF value of 11 m was exceeded.

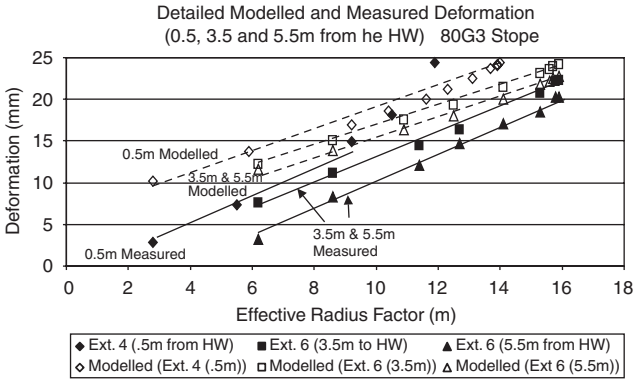


Fig. 12. Detailed measured deformation versus corresponding effective radius factor values measured 0.5, 3.5 and 5.5 m from the hanging wall contact 80G3 bench stope, Mount Isa Mines.

Fig. 12 shows the more detailed modelled and recorded deformation data for the anchors 0.5, 3.5 and 5.5 m from the hanging wall. In general, the magnitude of modelled deformations were greater than the field deformations, possibly due to the difficulty in estimating deformation properties for a rock mass. In addition, the estimated stress state for the stope is based on measurements of the virgin stress field. Near by mining activity will have changed the stress state in the area of stope 80G3. A more accurate estimate of the deformation properties of the rock mass, as well as the pre-mining stress state, may improve the match between the modelled and recorded deformation. A parametric study could be conducted to estimate the stresses and the rock mass deformation to more closely match the field deformation values. The purpose of this study is not to back analyse modelling results to match the measured value.

8. M665 Stoping Triplet

The M665 stoping triplet is a group of three stopes—M662, M665 and M667 as shown in Fig. 13. A detailed instrumentation study was carried out to record the behaviour of the rock and backfill within a stoping block to assist in the establishment of design principles. This work was summarised in internal Mount Isa Mine documents. The group of stopes is located at a depth of approximately 670 m with a stress of approximately 30 MPa normal to the ore body. Drilling and blasting for the open stopes were conducted from the overcut, undercut and from 3 sub-levels. The mining sequence consisted of Mining M667 followed by M662. During the mining of M662, M667 was backfilled with cemented hydraulic fill. M665 was then mass blasted into the open M662 stope. Maximum open stope dimensions were about 105 m vertical and 55 m on strike (stopes M662 and M665 together).

UNCORRECTED PROOF

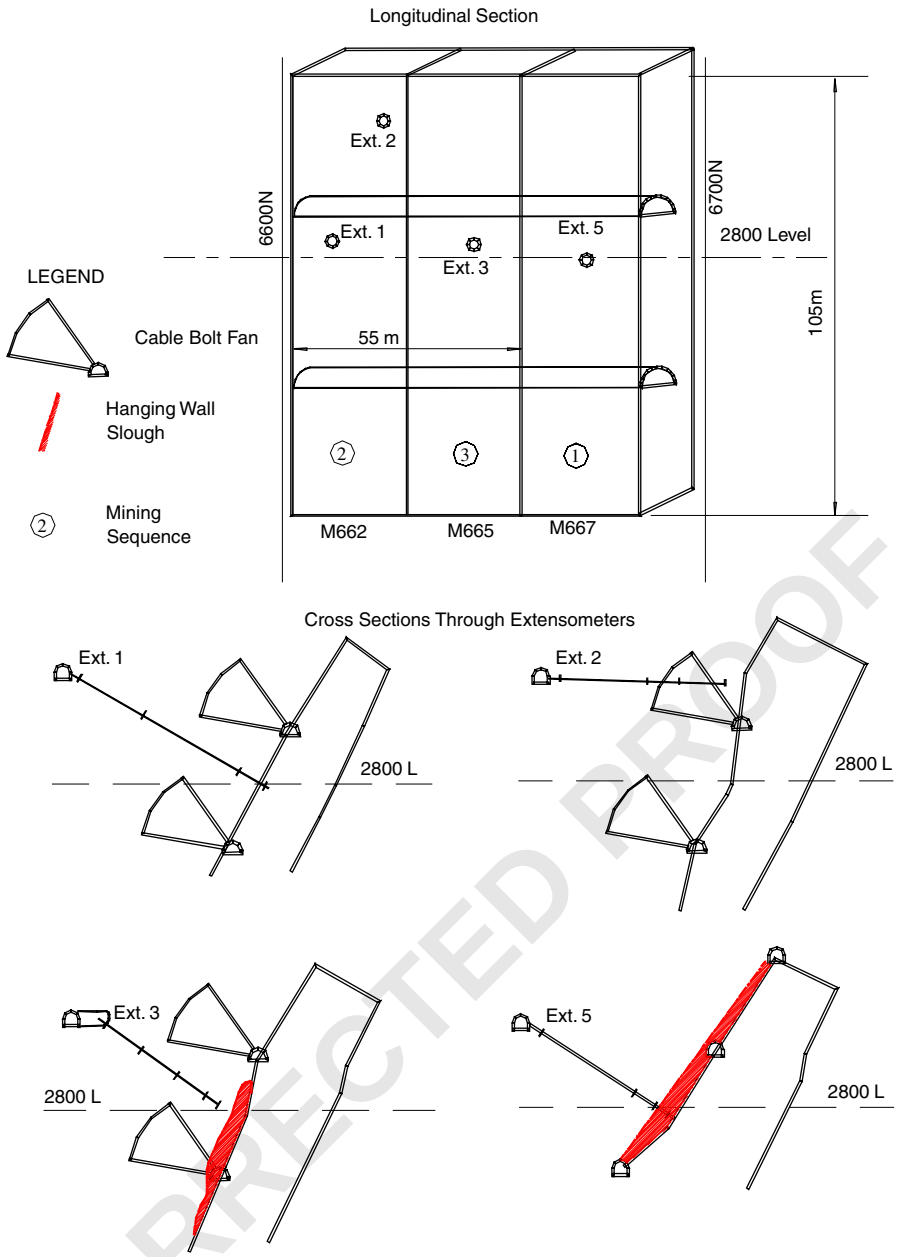


Fig. 13. Hanging wall instrumentation for M665 stoping block (after Milne [1]).

This instrumented stope study included 9 extensometers, a rock noise event counter, as well as CSIRO overcoring conducted at two separate stages of mining. The instrumentation of interest in this study are the extensometers located in the stope hanging wall recording deformation due to mining. Four extensometers, 1–3 and 5 were located in the stope hanging wall and recorded deformation due to mining past the instrument anchors. Extensometer 3 was located above stope M665 and was undercut only after the mass blasting of this stope resulting in only one ERF value and corresponding deformation. The backfilled abutment of stope M667 was treated as a rock abutment for calculating ERF values. Fig. 13 shows the location of these instruments on longitudinal and cross section views. The deformation between the collar and the anchor closest to the hanging wall was averaged for each major blast. This movement was plotted against the ERF value (Fig. 14). Data from extensometers 1–3 were plotted together since these instruments monitored a zone of rock between 4 and 8 m from the hanging wall.

Extensometers 1–3 monitor deformation 4–8 m from the open hanging wall. Extensometer data from both stopes M665 and M667 are shown together on this graph since they form a single opening after the mass blast of stope M665. The deformation initially showed a

UNCORRECTED PROOF

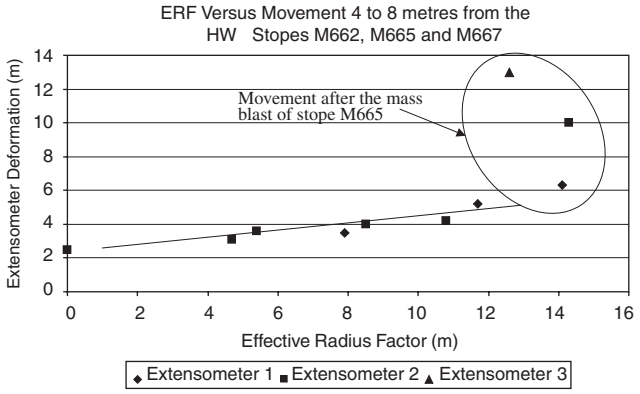


Fig. 14. Measured deformaiton versus corresponding effective radius factor values measured in the hanging wall for the M665 stoping triplet, 4–8 m from the hanging wall Mount Isa Mines (after Milne [1]).

linear increase in deformation with increasing ERF values (Fig. 14). This linear relation is similar to that observed with elastic numerical modelling. After the mass blast of stope M665, movements exceeded the values corresponding to this linear trend. The surrounding rock mass may have been weakened by the mass blast with the increased deformation rate indicating non-elastic deformations. It is interesting to note that after the mass blast, extensometer 1 movement is still close to the linear trend observed prior to the blast and extensometer 3 has diverged the most from this linear trend. Extensometer 1 is located furthest from the mass blast whereas extensometer 3 is located directly in the hanging wall of the mass blasted stope.

9. Conclusions

A limited amount of 3-D boundary element computer modelling has suggested a near linear relation exists between increasing ERF values on a surface and elastic deformation at a set distance above the surface. The magnitude and rate of movement with increasing ERF values will, however, vary with the stress field and elastic properties of the rock mass. This near linear relation between increasing ERF values and deformation above a surface has been observed with field data from two stoping areas at the Mount Isa Lead Mine. In the first case the linear relation between deformation and the ERF value was matched to numerically modelled deformation values. A similar linear trend was observed for both the field data and the modelling. In the second case, the relation between movement and ERF deviated from linearity after a mass blast, indicating the elastic deformation limit had been exceeded.

Field data from instruments such as extensometers give the engineer an excellent method of estimating the reaction of the rock mass to mining activity. The major drawback to using extensometer data is the fact that these instruments provide only a local assessment of rock mass behaviour. The ERF value has been used to generate a link between local measured deformation and the position on an opening surface. This allows the local measured rock mass deformation to be extrapolated over the total opening surface. Support design and pillar placement can be optimized based on the expected level of deformation linked to the surface geometry, defined by the ERF value [3], [12].

Acknowledgements

The authors wish to thank Xstrata Mount Isa Mines Limited for permission to publish this paper. The contribution of the Natural Sciences and Engineering Research Council of Canada (NSERC) towards this research is also acknowledged.

References

[1] Milne D. Underground design, deformation base on surface geometry, PhD thesis, The University of British Columbia, 1997. p. 251.
[2] Milne D, Pakalnis R, Lunder P. Approach to the quantification of hanging-wall behaviour. Trans Inst Min Metall 1996;105:A69–74.
[3] Milne D, Pakalnis R, Felderer M. Surface geometry assessment for open stope design. North American Rock Mechanics Symposium Montreal: 1996; p. 315–22.
[4] Potvin Y, Hudyma MR, Miller HDS. Design guidelines for open stope support. CIM Bulletin, June 1989.
[5] Wiles T. Issues of ease of use, error and accuracy control in Map3D mining analysis program in 3-dimensions. In: Poulin R, Pakalnis R, Mular AL, editors. Proceedings of the Second Canadian Conference on Computer Applications in the Mineral Industry. Vancouver, Canada: 1991. p. 669–77.
[6] Pascoe M, Oddie M. Assessing stope overbreak at Olympic Dam. In: Hebblewhite B, editor. Proceedings of the First Australasian Ground Control in Mining Conference. Sydney, Australia: 2003. p. 71–9.
[7] Windsor C, Bywater S, Worotnicki G. Instrumented and observed behaviour of N663 trial stope. Race Course Area, Mount Isa Mines, Project Report No. 23, CSIR Report. 1984. 54pp.
[8] Tucker G, Herbert D, Robinson J. Bench Stoping at Mount Isa Mines Limited Isa Lead Mine—Mount Isa, Queensland. Underground Operators' Conference. Townsville, Australia: 1998.
[9] Bywater S, Cowling R, Block N. Stress measurement and analysis for mine planning. Proceedings of the Fifth ISRM Congress. Melborne, Australia: 1983.
[10] Villaescusa E. Excavation design for bench stoping at Mount Isa Mine, Queensland, Australia. Trans Instn Min Metall London 1996; A1–10.
[11] Bieniawski Z. Determining rock mass deformability—experiences from case histories. Int J Rock Mech Min Sci Geomech Abstr 1978;15:237–47.
[12] Milne D, Hyett A, Bawden W, Pakalnis R. Interpreting deformation data for support optimization. Canadian Institute of Mining and Metallurgy, 100th Annual General Meeting. Montreal, Canada: 1998.

UNCORRECTED PROOF